Def.

Let $f:[a,b]\to\mathbb{C}$ be a function. Given a partition $P=\{x_0,\cdots,x_n\}$, define

$$V(f,P)\overset{def}{=}\sum_{i=1}^{n}|f(x_i)-f(x_{i-1})|$$

which is called the variation of $f$ with respect to a partition $P$. And, the value

$$TV(f)\overset{def}{=}\sup\{V(f,P)\mid P \text{ is Partition on } [a,b]\}$$

is called the total variation of $f$

If the total variation is finite, we say $f$ is of bounded variation.

Prop.

If $f:[a,b]\to\mathbb{C}$ is continuously differentiable, then $f$ is of bounded variation with

$$TV(f)=\int_a^b|f'(t)|\,dt$$

Proof. Given a partition $P=\{x_0,\cdots,x_n\}$ on $[0,1]$, by the fundamental theorem of calculus,

$$V(f,P)=\sum_{i=1}^{n}|f(x_i)-f(x_{i-1})|=\sum_{i=1}^{n}\left|\int_{x_{i-1}}^{x_i}f'(t)\,dt\right|\le\sum_{i=1}^{n}\int_{x_{i-1}}^{x_i}|f'(t)|\,dt=\int_a^b|f'(t)|\,dt$$

Since $P$ was arbitrary, $TV(f)\le\int_a^b|f'(t)|\,dt$, so the total variation has finite value.

To show that the reverse inequality, we use the fact that:

$f'$ is continuous uniformly, because it is continuous on the compact set $[a,b]$.

Given $\varepsilon>0$, take $\delta_1>0$ such that $|x-t|<\delta_1\Rightarrow|f(x)-f(t)|$.

And, take $\delta_2>0$ such that if a partition $P$ on $[a,b]$ satisfies $\|P\|<\delta_2$, then

$$\left|\int_a^b|f'(t)|\,dt-\sum_{i=1}^{n}|f'(z_i)|\cdot|x_{i+1}-x_i|\right|<\varepsilon$$

where $P=\{x_0,\cdots,x_n\}$ and $z_i\in[x_i,x_{i+1}]$ is arbitrary. The existence of $\delta_2$ is given by the fact that $|f'|$ is Riemann-integrable.

Prop. Let $f:[a,b]\to\mathbb{R}$ be a function. Given $c\in(a,b)$,

$$TV_a^b(f)=TV_a^c(f)+TV_c^b(f)$$

Proof. Given partitions $\mathbb{P}$ on $[a,c]$ and $\mathbb{Q}$ on $[c,b]$, the union $\mathbb{P}\cup\mathbb{Q}$ is a partition on $[a,b]$. Therefore, the equality given by definition

$$V_a^c(f,\mathbb{P})+V_c^b(f,\mathbb{Q})=V_a^b(f,\mathbb{P}\cup\mathbb{Q})\leq TV_a^b(f)$$

holds. Since $\mathbb{P}$ and $\mathbb{Q}$ were arbitrary,

$$TV_a^c(f)+TV_c^b(f)\leq TV_a^b(f).$$

To obtain the reverse inequality, let $\mathbb{B}$ be a partition on $[a,b]$. Then,

$$\begin{aligned}V_a^b(f,\mathbb{B})&\leq V_a^b(f,\mathbb{B}\cup\{c\})=V_a^c(f,(\mathbb{B}\cup\{c\})\cap[a,c])+V_c^b(f,(\mathbb{B}\cup\{c\})\cap[c,b])\\&\leq TV_a^c(f)+TV_c^b(f).\end{aligned}$$

Since $\mathbb{B}$ was arbitrary,

$$TV_a^b(f)\leq TV_a^c(f)+TV_c^b(f)$$

So, proved. ∎

Directly, $V(x)=TV_a^x(f)$ is well-defined monotone increasing function, if $f\in BV$.

Prop. Let $f:[a,b]\to\mathbb{R}$ be a bounded variation.
Then, for any $x\in[a,b)$, the right-sided limit $f(x+)$ exists.
Similarly, for any $x\in(a,b]$, the left-sided limit $f(x-)$ exists.
Proof. Since $f$ is of bounded variation, a map

$$V:[a,b]\to\mathbb{R}: t\mapsto TV_a^t(f)$$

is well-defined. Let $x\in[a,b)$ be given. Then, given $x<t_1<t_2\le b$,

$$|f(t_2)-f(t_1)|\le TV_{t_1}^{t_2}(f)=V(t_2)-V(t_1)$$

Let $\{x_n\}$ be a sequence in $(x,b]$ converging to $x$.
Since $V$ is monotone increasing, $V(x+)=\inf\{V(t)\mid t>x\}$ exists,
therefore a sequence $\{V(x_n)\}$ converges to $V(x+)$.
Now, given $\varepsilon>0$, there are $N_1, N_2\in\mathbb{N}$ such that
$n\ge N_1 \Rightarrow |V(x_n)-V(x+)|<\frac{\varepsilon}{2}$ and $m\ge N_2\Rightarrow |V(x_m)-V(x+)|<\frac{\varepsilon}{2}$
so $m,n\ge\max\{N_1,N_2\}$ implies

$$|V(x_n)-V(x_m)|\le|V(x_n)-V(x+)|+|V(x+)-V(x_m)|<\varepsilon$$

and since $|f(x_n)-f(x_m)|\le|V(x_n)-V(x_m)|$, $\{f(x_n)\}$ is also a Cauchy sequence,
so the limit $f(x+)$ of $\{f(x_n)\}$ exists. Since the sequence $\{x_n\}$ in $(x,b]$ was arbitrary, the right-sided limit $f(x+)$ exists.
The left-sided limit exists by the same way. $\square$

Note: But, the converse is not true, let

$$f:[0,1]\to\mathbb{R}: \begin{cases} 0 & \text{if } x=0 \\ x\cdot\sin\frac{1}{x} & \text{if } x\ne0. \end{cases}$$

Then, $f$ is <u>not</u> bounded variation, because: Consider the partition on $[0,1]$

$$P_n=\left\{0,1,\frac{2}{\pi},\frac{2}{3\pi},\cdots,\frac{1}{(2n+1)\pi}\right\}$$

$$\frac{1}{x}=\left(\frac{1}{2}+n\right)\pi$$